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RESEARCH ARTICLE

Expert Multi-Agent Conversational System Using Retrieval-Augmented Generation and Dynamic Text-to-SQL for Government Transparency

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ABSTRACT Access to structured and actionable public information remains a major challenge for government transparency in Peru, where official data are often fragmented, inconsistently published, and presented in low-quality formats. This study introduces a multi-agent conversational system designed under the Design Science Research paradigm, which is instantiated through three artifacts. First artifact is a data processing system that integrates procurement and legislative records through classification, region detection, and OCR. A second artifact is a set of specialized agents, combining Retrieval-Augmented Generation for attendance and voting with a Text-to-SQL agent for procurement. The third artifact is a multi-agent orchestration system that unifies these components within a microservices architecture. Together, they transform raw governmental records into structured, auditable, and citizen-accessible responses. The pipeline achieved strong results: classifiers reached accuracies up to 0.989, YOLOv11 obtained mAP@50–95 of 0.965 for attendance and 0.886 for voting, and Tesseract delivered a character error rate of 0.031 and word error rate of 0.078. The RAG agents achieved MAP of 0.950 for attendance and 0.926 for voting, confirming robust retrieval even under noisy conditions. For procurement queries, the Text-to-SQL agent outperformed baselines with FLEX of 0.909 and EX of 0.801. End-to-end evaluation confirmed reliability with Faithfulness of 0.942, Answer Relevancy of 0.907, Answer Similarity of 0.984, and Answer Correctness of 0.891, complemented by BERTScore, BLEU, and ROUGE metrics. These results validate the feasibility of large language models and microservice-based architectures to enhance transparency and accountability in low-data-quality environments.

INDEX TERMS Large language models (LLM), natural language processing (NLP), artificial intelligence, data integration, multi-agent systems, optical character recognition (OCR).

I. INTRODUCTION

Government transparency is a key component of democratic governance as it strengthens accountability, combats corruption, and promotes citizen participation. Internationally, various initiatives have promoted the use of digital technologies and open-data policies to facilitate access to public information. In Europe, countries like Estonia have led the implementation of interoperable systems that allow citizens to access real-time data on public services and administrative records [1]. However, setbacks persist in countries such as Hungary, Poland, and Austria, where restrictions on oversight bodies and corruption cases have been documented, undermining public trust [2], [3].

In Latin America, the landscape is similarly heterogeneous. According to the Open Data Barometer [4], Mexico, Uruguay, and Brazil lead in open data policies and digital government platforms, while a country such as Argentina has experienced regulatory backsliding, such as Decree 780/2024, which restricts access to public information [5]. At the other end of the spectrum are

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Venezuela and Nicaragua, where chronic institutional opacity persists [4], [6].

Beyond these cases, recent comparative studies highlight heterogeneous trajectories of open government across Latin America, including collaborative civic-tech platforms in Uruguay, variation in open data use by states and the Federal District in Brazil, and divergent municipal transparency patterns across 18 countries [7], [8], [9], [10].

In this context, the Peruvian case reflects a significant gap between the regulatory framework and its practical implementation [11]. The Law of Transparency and Access to Public Information (Law No. 27806) establishes clear obligations for the publication and delivery of data by state entities. However, various reports have documented systematic failures to comply with these provisions. For example, according to the Peruvian Office of the Ombudsman [12], between 2013 and 2022, over 57% of citizen complaints were due to the lack of response within the legal timeframe, and 31% of those were filed by people in vulnerable situations.

Furthermore, the fragmentation of institutional information and the lack of interoperability between platforms hinder unified access to information. Key data, such as public procurement, congressional debate records, purchase orders, and other administrative records, are dispersed across the portals of the Ministry of Economy and Finance (MEF), the Peruvian Government Procurement Supervisory Agency (OSCE), and the Congress of the Republic, among others.

The situation is worsened by the dominance of unstructured and non-reusable formats [13], [14], such as low-quality scanned documents, which hinder automatic information processing. Far from fostering practices aligned with open data principles, some public entities continue to generate and publish information under conditions that limit its effective use. This practice contributes to what the Organization for Economic Co-operation and Development (OECD) refers to as "Data Graveyards," that is, repositories that are publicly available but lack standard structures or adequate metadata, rendering them virtually unusable [15]. This situation is even more critical at the subnational level, where only 0.26% of regional and local governments have open data portals [14]; this scarce availability of structured and processable data not only severely limits citizen oversight and informed participation, but also constrains the development of advanced technological solutions.

To overcome these limitations, it is necessary to transform unstructured documents using techniques such as Optical Character Recognition (OCR), visual segmentation with models such as YOLO, and classification using Deep Learning (DL) models, enabling their integration into Retrieval-Augmented Generation (RAG) architectures and Large Language Models (LLMs).

Several studies have reported substantial improvements in accuracy and traceability when incorporating components for semantic retrieval and controlled generation [16], while international experiences in government automation show increases in operational efficiency and administrative transparency [17]. These achievements suggest high potential for applying similar technologies to governmental transparency initiatives.

In this study, we present a comprehensive multi-agent conversational system designed to improve governmental transparency in Peru through the intelligent integration of public information and natural language interfaces. The system addresses three critical domains: (1) public procurement records for companies with contracts above S/. 250,000, (2) congressional attendance records, and (3) voting records from 2006 to 2025.

Unlike prior studies focused on conceptual frameworks (Li et al. [18], who outline a five-component model for LLM-based multi-agent systems, and Guo et al. [19], who review coordination and communication challenges) and applied research such as Stein et al. [20], who employ a Design Science Research (DSR) approach to build a conversational agent for supporting data exploration and literacy in citizen science projects, or Papageorgiou et al. [21], limited to curated European Commission and OECD datasets, our research advances within low-data-quality settings by integrating heterogeneous scanned documents and spanning three governmental domains within a reproducible multi-agent architecture.

To address these challenges, the study follows the DSR paradigm through three complementary artifacts: (1) a Data Processing System for integrating procurement and legislative records via classification, region detection, and OCR; (2) a Set of Specialized Agents, including RAG-based agents for attendance and voting and a Text-to-SQL agent for procurement; and (3) a Multi-Agent Orchestration System that unifies them within a microservices architecture. Together, these artifacts transform raw governmental records into structured, auditable, and citizen-accessible answers.

II. RELATED WORK

This section surveys three technical pillars of our proposal: persistent challenges in open-government data, advances in document processing/OCR, and recent RAG methods for conversational access in public sector contexts.

A. OPEN GOVERNMENT DATA CHALLENGES

The complexity associated with open government data has been widely discussed in recent literature. Rodríguez et al. [22] analyzed over 102,000 public tenders in Spain and demonstrated that the lack of complete information about bidders limits system transparency and competitiveness, suggesting that better data quality could increase participation by up to 38% of tenders. Similarly, Lee and Jun [23] showed that attempts to integrate heterogeneous government datasets were hindered by the absence of unique identifiers, resulting in notably low matching rates.

On the other hand, the adoption of artificial intelligence in the government also reveals measurable benefits. The Chinese Smart Examination and Approval model has been extended to 31 out of 34 provinces and 129 cities, streamlining procedures and improving operational transparency through a combination of machine learning algorithms and auditable expert systems [17]. Likewise, a study in the United States found that 77% of public employees positively view the integration of AI into their administrative processes [24]. These results provide useful benchmarks for evaluating future government-transparency initiatives.

In Latin America, recent evidence emphasizes both opportunities and persistent gaps: municipal governments exhibit significant heterogeneity in transparency [10]; civic-tech initiatives such as Uruguay's Por Mi Barrio illustrate collaborative governance models [7]; and, in Brazil, there is uneven adoption of open government data by states and the Federal District [8], alongside differentiated transparency responses during the pandemic [9].

B. DOCUMENT PROCESSING AND OCR TECHNOLOGIES

Advanced document processing has significantly evolved to address the complexity of administrative files. Liu et al. demonstrated that integrating text and image analysis using deep neural networks and text detection with CTPN outperforms OCR-only approaches, achieving up to a 52% improvement in recall over the best traditional method [25].

Patil et al. applied semantic segmentation with U-Net on mixed-format documents and increased the Jaccard index from ~0.25 to ~0.69, representing nearly a 2.7× improvement in text-image matching [26].

YOLO architecture has also shown high effectiveness in document-related tasks. Pan et al. proposed WD-YOLO, which achieved a mAP@0.5 of 92.6% at 98 FPS thanks to contrast enhancement and dual attention modules [27]. In turn, the comprehensive review by Terven et al. details how different YOLO versions balance accuracy and speed, which is a key aspect for real-time processing of large volumes of documents [28].

C. RAG AND CONVERSATIONAL AI SYSTEMS

The combination of RAG with large language models is revolutionizing access to specialized information. In the medical domain, the interpretation accuracy of hepatology clinical guidelines rose from 43% to 99% when a RAG component was added to GPT-4 [16]. Similarly, a RAG-GPT system designed to advise on breast cancer nursing care increased evaluator nurse satisfaction to 8.4 ± 0.84, compared to 5.4 ± 1.27 from GPT-4 without external context ($p < 0.01$) [29].

Evaluation frameworks have also matured. RAGAS was initially introduced as a reference-free framework for assessing RAG systems, achieving ~0.95 faithfulness accuracy on the WikiEval dataset and outperforming GPTScore and GPT-ranking [30]. Since then, it has added metrics that accept reference answers, including Answer Relevancy, Answer Similarity, and Answer Correctness [31]. In addition, classical Natural Language Generation (NLG) metrics (BERTScore F1, BLEU, and ROUGE) are widely established and validated in the literature [32], [33].

The FC-MAS model, proposed in a recent state-of-the-art study on multi-agent systems, introduces a five-layer architecture that helps measure and mitigate communication overload and other bottlenecks in distributed implementations, although comparable metrics have not yet been published [34].

However, most advances in RAG and LLM evaluation have been concentrated in English. In Ibero-American contexts, dedicated pretrained models such as MarIA for Spanish [35], BERTimbau for Brazilian Portuguese [36], and RoBERTuito for Spanish social media [37] demonstrate that linguistic resources are maturing. Moreover, multilingual foundations such as XLM-R [38] enable cross-lingual transfer into Spanish and Portuguese tasks, underscoring the importance of adapting retrieval and generation pipelines to these languages.

Beyond linguistic resources, recent surveys on multi-agent systems further consolidate the relevance of modular system designs: Li et al. propose a five-component framework (profile, perception, self-action, interaction, and evolution) that can be interpreted as a modular architecture for LLM-based multi-agent systems [18]. Guo et al. complement this perspective by reviewing progress and open challenges in multi-agent coordination and communication [19]. Together, these studies underscore the importance of adopting distributed and component-based architectures, such as the one proposed in this work.

Given that our deployment is situated in the public sector, we require an architecture that is both modular and scalable. Evidence from regulated domains shows that microservice designs, combined with API gateways and governance practices such as versioning and backward compatibility, ensure interoperability while avoiding organizational lock-in [39], [40]. To guarantee methodological rigor, we follow the Design Science Research (DSR) paradigm, which provides a structured approach for building and evaluating artifacts in e-government contexts [20]. This orientation is supported by real implementations: Papageorgiou et al. demonstrated that modular RAG architectures can be integrated into public services while preserving transparency [21], and Stein et al. highlighted that DSR-based conversational agents facilitate citizen data exploration, reinforcing their scalability and value for government services [20].

III. DESIGN SCIENCE RESEARCH METHODOLOGY

This study adopts the DSR paradigm, which provides a structured process for designing, building, and evaluating technological artifacts to solve ill-structured problems. As outlined by Hevner and Chatterjee [41], Peffers et al. [42], and Gregor and Hevner [43], and more recently applied by Stein et al. [20] in the design of conversational agents for citizen science data exploration, DSR guides research

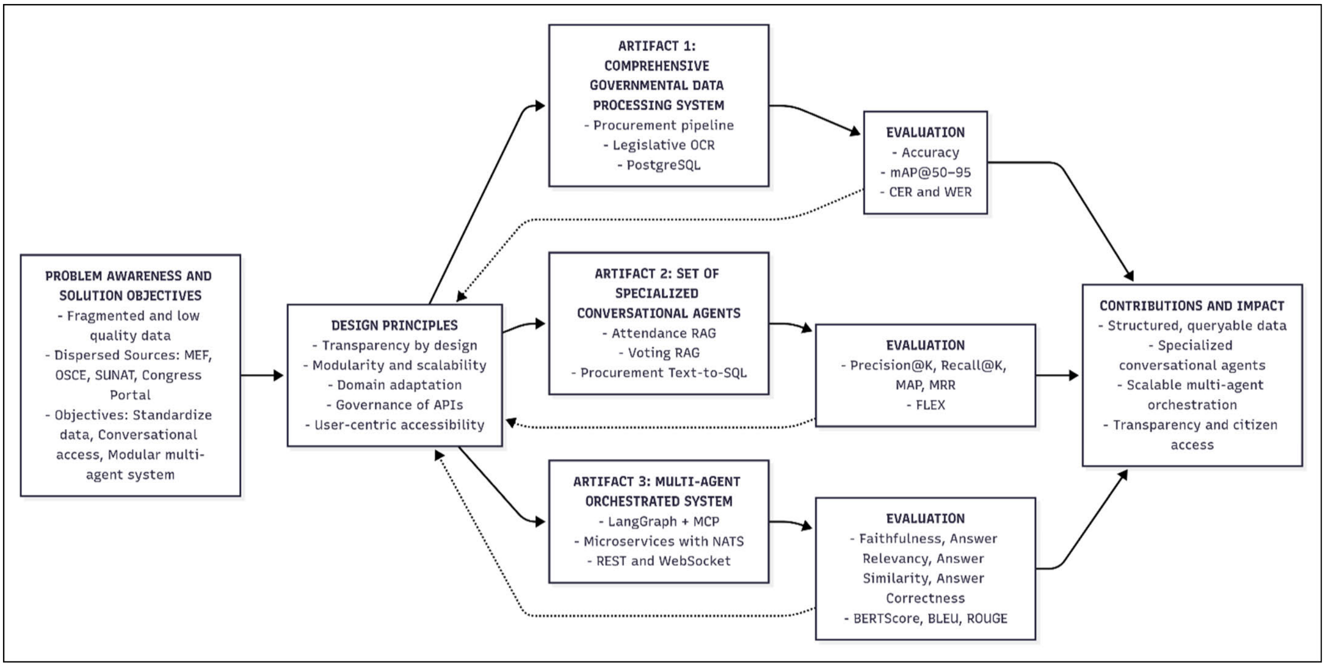

FIGURE 1. **Design Science Research workflow and artifacts.**

through iterative cycles of problem identification, objective formulation, artifact development, and rigorous evaluation.

A. PROBLEM AWARENESS AND SOLUTION OBJECTIVES

The design problem in this research is the fragmentation and low quality of Peruvian governmental data. Procurement records (MEF, OSCE, SUNAT) and legislative documents are dispersed and often published in non-standard formats, such as low-resolution scanned PDFs without metadata. These "data graveyards" are technically open but difficult to use, limiting transparency, citizen oversight, and technological innovation.

Accordingly, the design objectives are threefold:

1) Transform heterogeneous governmental datasets into standardized and queryable formats.
2) Enable domain-specific conversational access through specialized agents.
3) Integrate these agents into a modular and scalable multi-agent architecture suitable for public-sector applications.

B. DESIGN PRINCIPLES FOR GOVERNMENT CONVERSATIONAL AGENTS

To achieve these objectives, the solution is structured into three complementary artifacts:

- Transparency by design: Data pipelines must generate structured, auditable, and traceable outputs to support accountability.
- Modularity and scalability: The architecture must consist of independent services that can evolve and scale horizontally.
- Domain adaptation: Different types of governmental data require tailored processing and retrieval strategies (e.g., RAG for unstructured legislative texts, Text-to-SQL for structured procurement databases).
- Governance of APIs and data: Interfaces must follow principles of versioning, backward compatibility, and standardization to ensure long-term maintainability.
- User-centric accessibility: Natural-language interfaces must reduce barriers for citizens to access and query governmental information.

These principles establish the foundation for three complementary artifacts: (1) a Comprehensive Governmental Data Processing System, (2) a Set of Specialized Conversational Agents, and (3) a Multi-Agent Orchestrated System. Figure 1 illustrates the overall DSR process, showing these three artifacts, their interdependencies, and their evaluation strategy.

IV. ARTIFACTS AND EVALUATION

Building on the design objectives and principles defined in Section III, this section presents the three artifacts that constitute the proposed solution. For each artifact, we summarize its purpose, describe the design and evaluation metrics, and report the corresponding results. The collective contributions and impact of the three artifacts are synthesized in the discussion section (Section V).

A. ARTIFACT 1: DATA PROCESSING SYSTEM

1) PURPOSE

This artifact addresses the fundamental challenge of making fragmented and unusable Peruvian governmental data accessible and processable for conversational AI systems.

It specifically targets two critical problems documented in the literature: the dispersion of public procurement information across disconnected platforms (MEF, OSCE, SUNAT), which prevents comprehensive public spending analysis and limits citizen oversight, and the publication of congressional legislative documents as scanned images without searchable text—what the OECD calls "data graveyards" [15]. The purpose of this artifact is to transform these raw and heterogeneous resources into structured, queryable, and auditable datasets, thereby enabling the subsequent design of conversational agents that can provide evidence-based responses about governmental operations and reduce barriers to public information access.

2) DESIGN

This artifact implements dual-pipeline architecture to process both structured and unstructured governmental data sources, orchestrated through containerized extraction services and specialized processing workflows.

a: DATA EXTRACTION COMPONENT

The data extraction process focused on Peruvian governmental platforms relevant to public procurement and legislative activities. Records were extracted from the Economic Transparency Portal for state supplier companies with accumulated contracts exceeding S/. 250,000, identifying 499,488 companies. Subsequently, 12,677,736 contracts and service orders associated with these companies were retrieved from the OSCE API.

To complement this information, 452,432 records were extracted from the portal of the National Superintendency of Tax Administration (SUNAT), including key data on the incorporation date and tax status of suppliers. For Congress-related data, 938 PDF files containing 16,376 pages with attendance and voting records spanning 2006 to 2025 were used.

All data were collected up to May 17, 2025, which constitutes the official cut-off date for the dataset. Figure 2 presents the general extraction scheme applied to each source, highlighting the use of scrapers and APIs, depending on the technical nature of each platform.

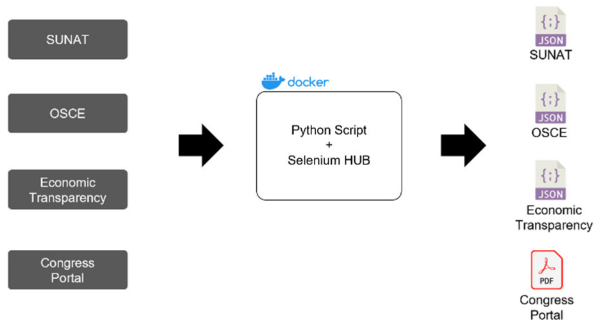

FIGURE 2. Extraction workflow for each data source. The process combines automated scrapers and API calls, orchestrated through a Dockerized Selenium HUB environment, producing datasets in JSON and PDF formats.

TABLE 1. Configurations for classification models.

Column Name	Description
ruc_proveedor	Unique taxpayer ID of the supplier
nombre_proveedor	Legal name of the supplier
monto_girado	Total amount disbursed by the government
tipo_empresa	Classification as legal entity or natural person
tipo_contribuyente	Taxpayer category
fecha_inicio_activi dades	Official start date of the supplier's economic activity

TABLE 2. Schema of contracts_orders table.

Column Name	Description
ruc_proveedor	Unique taxpayer ID of the supplier
nombre_proveedor	Legal name of the supplier
entidad_contratante	Contracting government agency
objeto_contrato	General object of the contract
descripcion	Detailed description of the contract
moneda	Currency used in the contract
monto	Total amount awarded
fecha_inicio	Contract start date
fecha_fin	Contract end date
tipo_contrato	Contract type

b: STRUCTURED DATA PIPELINE

The collected data were transformed and filtered to retain only the most relevant columns for the development of conversational agents. Two main tables were constructed for complementary purposes. The transparency_sunat table (Table 1) enables viewing of the total amounts contracted by each state supplier, along with additional information on their tax and legal profile, providing a consolidated view per Unique Taxpayer ID (RUC). By contrast, the contract orders table (Table 2) offers a granular level of detail, including contract type, contracting entity, amounts, and execution dates, allowing specific analyses by contractual operation.

Both structures were loaded into a PostgreSQL database and optimized by indexing key fields, such as ruc_proveedor and fecha_inicio. This relational organization was designed to facilitate efficient querying by the specialized public procurement agent described in the Agent Design Section.

c: UNSTRUCTURED DATA PIPELINE

To transform legislative documents into analyzable inputs (Figure 3). Each stage addresses a different technical challenge: (1) image extraction, (2) document classification, (3) region of interest detection, and (4) OCR extraction with JSON structuring.

i) Image Extraction

The 938 PDFs downloaded from the Peruvian Congress portal were segmented into 16,376 images. The corpus

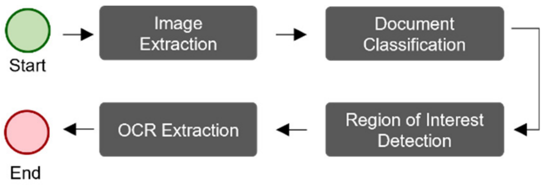

FIGURE 3. Unstructured data processing pipeline.

TABLE 3. Configurations for classification models.

Model	Learning Rate	Batch Size
DenseNet121	1e-4, 1e-5, 1e-6	32
EfficientNet-B0	1e-4, 1e-5, 1e-6	32
MobileNetV2	1e-4, 1e-5, 1e-6	32
ResNet50	1e-4, 1e-5, 1e-6	32
VGG16	1e-4, 1e-5, 1e-6	32

included attendance records, voting reports, and miscellaneous official documents (minutes, circulars, and formal notes). Each page was exported in the PNG format at 300dpi to preserve the resolution required in subsequent stages.

ii) Document Classification

To identify relevant documents automatically, a balanced dataset of 900 images was constructed and evenly distributed across three classes: attendance, voting, and others (e.g., minutes, circulars, or institutional notes). Figure 4 presents visual examples illustrating the diversity of these categories.

The dataset was split into 80% for training/validation, and 20% for testing, enabling model performance to be evaluated under reproducible conditions. To strengthen statistical validity, the training/validation portion was further assessed with stratified 10-fold cross-validation (CV), a widely adopted practice that balances bias and variance in accuracy estimation [44], [45]. The Reported results correspond to the mean accuracy across folds with 95% Wilson score confidence intervals [46], [47], providing bounded and interpretable estimates of variability, particularly suitable for binomial outcomes.

A transfer learning strategy was applied using pretrained models on ImageNet and adapting them to the task by replacing their final layer with a fully connected linear layer with three nodes corresponding to the defined classes.

The evaluated configurations included combinations of architecture, learning rates, and batch sizes, as detailed in Table 3. All models were trained for 20 epochs using PyTorch on the GPU, monitoring performance on the validation set to select the best classifier.

The selected model was then applied to 16,376 images to filter only attendance and voting documents that were processed in subsequent stages.

iii) Region of Interest Detection

Based on the classification stage, 6,830 attendance documents and 8,989 voting documents were segmented visually.

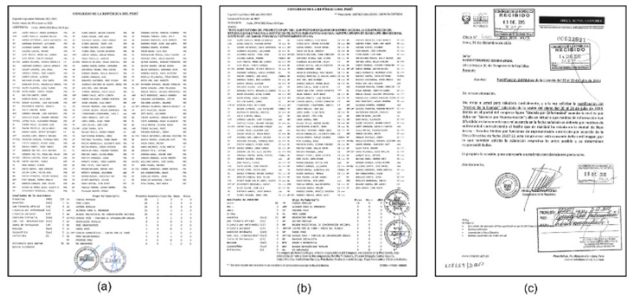

FIGURE 4. Representative examples by document type: (a) attendance, (b) voting, (c) other.

To train models capable of automatically locating key areas in each document type, 400 images were manually annotated and equally divided between classes (200 per type). The labeled regions included headers, name columns, legislative groups, and totals in attendance documents and headers, subject, session president, voting results, and signatories in voting documents. Figure 5 shows representative examples of these annotations.

Although only the header region was used in the following stages for metadata extraction, additional areas were labeled to provide more context to the model during training. This strategy improves the accuracy of header detection, which is critical for correctly indexing documents into the knowledge base.

YOLOv11, which is a modern architecture optimized for detection tasks with high visual variability, was selected for this task. Its design incorporates an efficient backbone and advanced techniques, such as anchor-free detection and token pruning, which improve both accuracy and inference speed [48], [49]. These features make YOLOv11 particularly suitable for detecting semantic regions in variable-quality scanned documents.

Two independent YOLOv11 models were trained: one for attendance documents and the other for voting documents. For each case, different combinations of architecture, image size, and batch size were evaluated with a maximum duration of 50 epochs and validation at each iteration. An early stopping criterion with patience of 10 epochs was employed.

Model performance was estimated using 5-fold CV to reduce bias from a single split [44]. Although 10-fold CV is often considered standard, prior work shows that 5- and 10-fold provide comparable estimates [45], [50]. Given the high cost of training YOLO, 5-fold CV offers a practical compromise. Reported results correspond to the mean mAP@50–95 with 95% confidence intervals from the Student's t-distribution, which is appropriate for continuous fold-level metrics under small sample sizes (n=5). The combinations explored are summarized in Tables 4 and 5.

iv) OCR Extraction

Once the regions of interest were identified, header region was cropped in attendance documents, while the header, subject, and session president were cropped in voting documents,

TABLE 4. Parameter combinations explored for attendance documents.

Model	Image Size (pixels)	Batch Size
yolo11	320, 480, 640	16, 32
yolo11n	320, 480, 640	16, 32
yolo11s	320, 480, 640	16, 32
yolo11x	320, 640	16, 32

TABLE 5. Parameter combinations explored for voting documents.

Model	Image Size (pixels)	Batch Size
yolo11	320, 480, 640	16, 32
yolo11n	320, 480, 640	16, 32
yolo11s	320, 480, 640	16, 32
yolo11x	320, 640	16, 32

FIGURE 5. Representative examples of manually annotated documents by type: (a) attendance record, (b) voting report. Colored regions indicate semantically distinct zones used for training the YOLOv11 models.

as these areas concentrated the most relevant metadata, such as the legislature, session date, and document type. Cropping not only isolates essential information for later indexing but also improves OCR performance by reducing noise and visual ambiguity from broader regions [25], [26].

To determine the most suitable OCR engine, four widely used alternatives were compared: Tesseract, EasyOCR, Doctr, and PaddleOCR. In line with previous OCR research, we adopted the Character Error Rate (CER) and Word Error Rate (WER) as evaluation metrics, since they are the standard measures for text recognition performance [51], [52]. The evaluation was performed on a test set composed of 400 manually annotated images across four region types (Table 6), totaling 49,460 characters and 7,602 words (100 images per category).

We applied a percentile bootstrap resampling procedure with 2,000 iterations to compute 95% confidence intervals for CER and WER. This approach is appropriate because

TABLE 6. Distribution of manually annotated dataset.

Region	Number of Images	Characters	Words
Subject	100	22,036	3,591
Attendance header	100	11,741	1,724
Voting header	100	11,541	1,718
Session president	100	4,142	569
Total	400	49,460	7,602

error rates are bounded, often skewed, and do not follow a normal distribution, which makes classical parametric assumptions unreliable. Bootstrap intervals provide robust, distribution-free estimates of uncertainty. Although prior studies have focused on NLP and text evaluation, the same rationale applies to character- and word-level error rates in OCR [32], [33], [53]. All tests were conducted under consistent conditions—300 dpi resolution and no additional preprocessing—to measure the native performance of each engine.

The extracted text was normalized using Unicode conversion (NFKC), newline removal, and correction of common errors, and then structured into JSON files containing the identified key fields. This format facilitates integration into the downstream indexing and retrieval workflows.

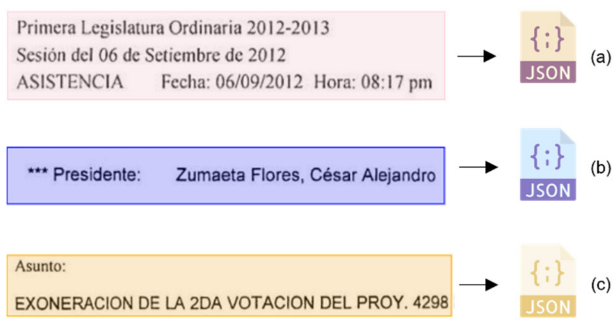

FIGURE 6. Representative cropped regions used for OCR: (a) header in an attendance document, (b) session president in a voting document, and (c) subject in a voting document.

3) METRICS

i) Image Classification

The automatic classification of legislative documents was evaluated using an accuracy metric, which is defined as the proportion of correct predictions over the total number of examples. It is calculated as:

$$\text{accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \times 100\% \tag{1}$$

where TP and TN represent true positives and true negatives, respectively, and FP and FN correspond to misclassification. This metric is appropriate in balanced contexts such as ours, and is widely used in visual recognition tasks, including the ImageNet ILSVRC challenge [54].

ii) Object Detection

The performance of the YOLOv11 models in detecting key regions within legislative documents was evaluated using the Mean Average Precision at multiple thresholds (mAP@50–95). This metric has become the standard evaluation measure in modern benchmarks, such as MS COCO, owing to its ability to simultaneously capture classification accuracy and spatial localization quality [55].

Unlike more lenient metrics such as mAP@0.5, the mAP@50–95 value is obtained by averaging the Average Precision (AP) across ten Intersection over Union (IoU) thresholds, ranging from 0.50 to 0.95 in increments of 0.05. Formally, it is defined as:

$$\text{mAP@50}-95 = \frac{1}{10}\sum_{t=0.50}^{0.95}\left(\frac{1}{|C|}\sum_{c\in C}\text{AP}_c^{(t)}\right) \tag{2}$$

where $\text{AP}_c^{(t)}$ denotes the average precision for class c at IoU threshold t, and C is the set of considered classes. This approach encourages models that not only detect the correct classes but also do so with high spatial precision under various scenarios.

The exclusive use of mAP@50–95 was adopted because of its more rigorous and discriminative nature [56]. In the context of this study, its application is particularly relevant given the high degree of visual and structural variability presented in scanned documents.

iii) Optical Character Recognition

The accuracy of the OCR engine was evaluated using two fundamental metrics in the field of text recognition: CER and WER. Both metrics are derived from the Levenshtein distance, which quantifies the minimum number of operations required to transform one sequence into another by considering substitutions (S), deletions (D), and insertions (I).

The character error rate is defined as:

$$\text{CER} = \frac{S+D+I}{N} \tag{3}$$

where N is the total number of characters in the reference text.

On the other hand, the word error rate is calculated as:

$$\text{WER} = \frac{S+D+I}{N_w} \tag{4}$$

where N_w represents the total number of words in the reference transcription. This metric is particularly well suited for structured documents, such as administrative headers or official records, where lexical consistency is the key to interpretation.

These metrics have been widely used in the evaluation of optical character recognition and automatic speech recognition systems and are considered a standard in the technical literature [57].

TABLE 7. Cross-validated accuracy and test accuracy across 15 document classification model configurations.

Experiment	Validation	Test
densenet121_lr1e4_bs32	0.999 [0.996, 1.000]	0.989
densenet121_lr1e5_bs32	0.999 [0.996, 1.000]	0.989
densenet121_lr1e6_bs32	0.989 [0.980, 0.998]	0.944
efficientnetb0_lr1e4_bs32	0.999 [0.996, 1.000]	0.989
efficientnetb0_lr1e5_bs32	0.999 [0.996, 1.000]	0.978
efficientnetb0_lr1e6_bs32	0.953 [0.927, 0.980]	0.889
mobilenet_v2_lr1e5_bs32	0.995 [0.991, 1.000]	0.978
mobilenetv2_lr1e4_bs32	0.998 [0.994, 1.000]	0.989
mobilenetv2_lr1e6_bs32	0.924 [0.887, 0.960]	0.889
resnet50_lr1e4_bs32	0.999 [0.996, 1.000]	0.989
resnet50_lr1e5_bs32	0.999 [0.996, 1.000]	0.978
resnet50_lr1e6_bs32	0.986 [0.979, 0.994]	0.933
vgg16_lr1e4_bs32	0.998 [0.994, 1.000]	0.989
vgg16_lr1e5_bs32	0.999 [0.996, 1.000]	0.978
vgg16_lr1e6_bs32	0.994 [0.988, 1.000]	0.978

4) RESULTS

i) Document Classification Performance

Fifteen deep-learning model configurations were evaluated for legislative document classification. Results are reported using the nomenclature model_lrX_bsY, where model indicates the architecture (e.g., ResNet50, DenseNet121), lrX the learning rate (e.g., lr1e4 = 1 × 10.4), and bsY the batch size (e.g., bs32 = 32).

The top-performing models were ResNet50, VGG16, and DenseNet121, which achieved cross-validated accuracies above 99% with narrow 95% confidence intervals (close to [0.996, 1.000]) and test accuracies up to 0.989, clearly outperforming alternatives such as EfficientNet and MobileNet. ResNet50 was selected as the final model because of its shorter training time while maintaining competitive performance.

These results are summarized in Table 7, which presents the accuracy values obtained under stratified 10-fold cross-validation (mean and 95% CI) together with test performance.

ii) Region of Interest Detection with YOLOv11

The automatic detection of key areas in attendance and voting documents was addressed through 32 experiments using YOLOv11 models. Results are reported using the nomenclature model_imgX_bsY, where model indicates the YOLO variant (e.g., yolo11n, yolo11s), imgX the input image size in pixels (e.g., img320 = 320 × 320), and bsY the batch size (e.g., bs32 = 32).

For attendance documents, several configurations achieved high performance, with most variants surpassing a mAP@50–95 of 0.94, and the yolo11n_img640_bs16 model exhibiting the best performance with 0.965 (95% CI: 0.962–0.969) (Table 8).

TABLE 8. Performance of YOLOv11 models for region of interest detection in attendance documents.

Experiment	mAP@50–95
yolo11n_img640_bs16	0.965 [0.962, 0.969]
yolo11n_img640_bs32	0.962 [0.956, 0.968]
yolo11s_img640_bs32	0.947 [0.939, 0.956]
yolo11n_img480_bs32	0.944 [0.918, 0.970]
yolo11s_img480_bs32	0.941 [0.921, 0.960]
yolo11s_img640_bs16	0.939 [0.910, 0.967]
yolo11n_img320_bs32	0.930 [0.906, 0.954]
yolo11s_img320_bs32	0.919 [0.912, 0.926]
yolo11l_img320_bs32	0.875 [0.836, 0.915]
yolo11l_img480_bs32	0.875 [0.774, 0.975]
yolo11l_img640_bs16	0.833 [0.772, 0.893]
yolo11l_img640_bs32	0.822 [0.761, 0.883]
yolo11x_img640_bs32	0.525 [0.396, 0.653]
yolo11x_img320_bs32	0.486 [0.225, 0.748]
yolo11x_img640_bs16	0.486 [0.311, 0.661]
yolo11x_img480_bs32	0.432 [0.192, 0.672]

TABLE 9. Performance of YOLOv11 models for region of interest detection in voting documents.

Experiment	mAP@50–95
yolo11n_img320_bs32	0.831 [0.812, 0.850]
yolo11n_img480_bs32	0.875 [0.868, 0.881]
yolo11n_img640_bs32	0.879 [0.866, 0.893]
yolo11n_img640_bs16	0.873 [0.858, 0.887]
yolo11s_img320_bs32	0.851 [0.827, 0.876]
yolo11s_img480_bs32	0.848 [0.757, 0.939]
yolo11s_img640_bs32	0.870 [0.847, 0.893]
yolo11s_img640_bs16	0.886 [0.873, 0.899]
yolo11l_img320_bs32	0.798 [0.712, 0.883]
yolo11l_img480_bs32	0.776 [0.762, 0.790]
yolo11l_img640_bs32	0.794 [0.711, 0.878]
yolo11l_img640_bs16	0.861 [0.761, 0.961]
yolo11x_img320_bs32	0.455 [0.291, 0.618]
yolo11x_img480_bs32	0.453 [0.259, 0.646]
yolo11x_img640_bs32	0.582 [0.498, 0.666]
yolo11x_img640_bs16	0.642 [0.552, 0.731]

For voting documents, the yolo11n_img640_bs32 model achieved a mAP@50–95 of 0.879 (95% CI: 0.866–0.893), with the majority of variants scoring above 0.85 (Table 9).

The use of 5-fold cross-validation ensured that the reported performance metrics are not dependent on a single train/validation split, but instead reflect stable and statistically reliable estimates.

Both sets of results confirm the robustness of the adopted approach for semantic document segmentation, which is a crucial step for subsequent processes, such as OCR and structured data extraction.

TABLE 10. Comparison of OCR engines by CER and WER.

OCR	Region	CER	WER
Tesseract	Subject	0.029 [0.015, 0.046]	0.064 [0.038, 0.094]
	Attendance header	0.009 [0.007, 0.011]	0.029 [0.023, 0.035]
	Voting header	0.004 [0.002, 0.005]	0.013 [0.008, 0.018]
	Session President	0.080 [0.044, 0.122]	0.204 [0.144, 0.271]
	All	0.031 [0.020, 0.042]	0.078 [0.060, 0.098]
EasyOCR	Subject	0.028 [0.019, 0.038]	0.088 [0.068, 0.109]
	Attendance header	0.022 [0.017, 0.028]	0.112 [0.091, 0.134]
	Voting header	0.026 [0.019, 0.034]	0.127 [0.101, 0.157]
	Session President	0.108 [0.082, 0.138]	0.308 [0.258, 0.361]
	All	0.046 [0.038, 0.055]	0.159 [0.141, 0.178]
Doctr	Subject	0.038 [0.029, 0.047]	0.139 [0.120, 0.159]
	Attendance header	0.010 [0.009, 0.012]	0.062 [0.059, 0.066]
	Voting header	0.015 [0.014, 0.016]	0.097 [0.090, 0.104]
	Session President	0.096 [0.058, 0.140]	0.277 [0.192, 0.375]
	All	0.040 [0.030, 0.051]	0.144 [0.121, 0.170]
PaddleOCR	Subject	0.197 [0.152, 0.246]	0.459 [0.411, 0.507]
	Attendance header	0.050 [0.045, 0.056]	0.407 [0.363, 0.450]
	Voting header	0.060 [0.053, 0.067]	0.430 [0.384, 0.475]
	Session President	0.209 [0.154, 0.270]	0.734 [0.677, 0.788]
	All	0.129 [0.110, 0.150]	0.507 [0.480, 0.534]

iii) Performance OCR Evaluation

Four OCR engines were compared using CER and WER with 95% confidence intervals estimated through bootstrap resampling (2,000 iterations).

As summarized in Table 10, Tesseract achieved the best overall performance, with an average CER of 0.031 [0.020, 0.042] and WER of 0.078 [0.060, 0.098]. Its advantage was most evident in the attendance and voting headers, where it reached error rates an order of magnitude lower than EasyOCR and PaddleOCR.

The most challenging case was the session president region, where all engines exhibited notably higher error rates. This was mainly due to the presence of stamps overlapping the text and the lower quality of the scanned images, which added significant visual noise. Representative examples of these issues are shown in Figure 7. Despite these difficulties, Tesseract remained the most reliable option, supporting its adoption in the final pipeline.

FIGURE 7. Representative examples of session president regions affected by overlapping stamps.

B. ARTIFACT 2: SPECIALIZED CONVERSATIONAL AGENTS

1) PURPOSE

This artifact focuses on the design of specialized agents that act as intermediaries between the processed datasets from Artifact 1 and the orchestration system defined in Artifact 3. Their purpose is to retrieve structured or semi-structured information from governmental sources and return it as consistent and auditable outputs, without yet performing natural language generation. Each agent adapts to the nature of its dataset:

- **Procurement Agent**, based on Text-to-SQL techniques, translates natural-language queries into SQL statements over the PostgreSQL database and returns the corresponding results.
- **Attendance Agent**, leveraging a RAG approach, retrieves OCR-processed passages from legislative attendance documents.
- **Voting Agent**, also RAG-based, extracts evidence about voting results, including OCR text and metadata.

Through this configuration, the artifact ensures that each governmental domain is equipped with a dedicated retrieval agent capable of delivering reliable outputs, which are later integrated by the orchestration system (Artifact 3) through the MCP protocol.

2) DESIGN

The design of this artifact is organized around two technical approaches. The first approach corresponds to agents that rely on RAG for accessing OCR-based legislative documents (Attendance and Voting Agents). The second approach corresponds to the Text-to-SQL paradigm, applied by the Procurement Agent to operate over the relational database. The following subsections detail the specific architecture, components, and interaction flows of each approach.

a: ATTENDANCE AND VOTING AGENTS (RAG)

The attendance and voting agents share a RAG architecture, designed to operate on legislative documents previously processed by OCR. The indexing step is performed only once during the preparation phase: each document was segmented into overlapping chunks of 512 tokens and vectorized using OpenAI's text-embedding-3-small model. The resulting vectors were stored in Qdrant, a vector database that enables fast and scalable semantic search.

During the operational phase, the agents do not repeat the indexing; instead, they follow a minimal flow focused exclusively on retrieval (Figure 8):

1) **Input**: the agent receives the query.
2) **Call Qdrant**: the query is converted into an embedding and sent to the vector database.
3) **Chunk retrieval**: the most relevant segments are selected based on semantic similarity.
4) **Output**: the retrieved fragments are returned in structured format through the MCP protocol.

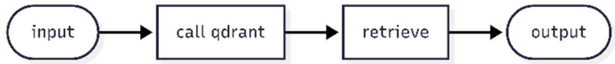

FIGURE 8. Internal flow of the attendance and voting agents (RAG).

b: PUBLIC PROCUREMENT AGENT (TEXT-TO-SQL)

The Public Procurement Agent operates on the *transparency_sunat* and *contracts_orders* tables, which were built in Artifact 1 and loaded into PostgreSQL. Its role is to transform natural-language questions into dynamic SQL statements and return consistent and auditable tabular outputs.

The internal flow, illustrated in Figure 9, follows these main steps:

1) **Input**: the agent receives the query in natural language.
2) **List tables**: the available tables in the database are identified.
3) **Get schema**: columns and data types are retrieved through a schema call.
4) **Generate query**: a candidate SQL statement is generated based on the database structure and query intent.
5) **Check query**: the generated SQL is verified for syntactic and semantic validity.
6) **Run query**: the validated query is executed in PostgreSQL.
7) **Output**: tabular results in structured format via MCP.

This design ensures precision, traceability, and consistency when accessing the structured datasets derived from Artifact 1.

3) METRICS

i) RAG agents

For the attendance and voting agents, a set of classical information retrieval metrics was adopted, which recent literature recognizes as appropriate for assessing the quality of the search module within RAG architectures [58] The formal definitions are presented below.

First, Precision@K quantifies the proportion of relevant fragments among the top K returned and is formalized as:

$$\text{Precision@}K = \frac{1}{K}\sum_{k=1}^{K} rel(k) \tag{5}$$

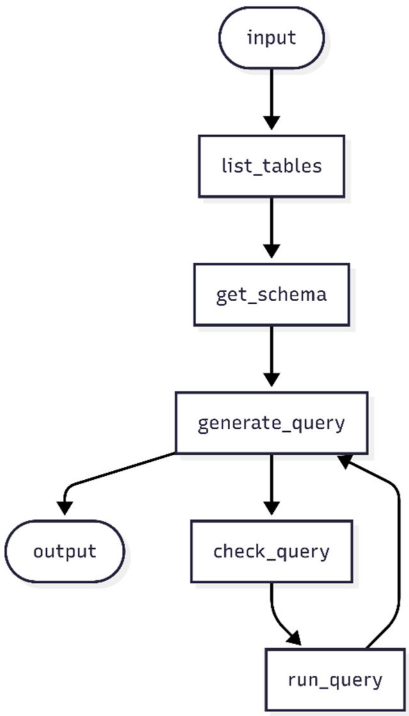

FIGURE 9. Internal flow of the public procurement agent (Text-to-SQL).

where $rel(k) = 1$ if the document at position k is relevant and 0 otherwise. Complementarily, Recall@K measures the retrieval coverage over the total set of relevant documents:

$$\text{Recall@}K = \frac{1}{|R|}\sum_{k=1}^{K} rel(k) \tag{6}$$

with $|R|$ equal to the number of known relevant fragments for the query.

To simultaneously weigh the precision and ranking, the Mean Average Precision is used:

$$\text{MAP} = \frac{1}{|Q|}\sum_{q\in Q}\frac{1}{|R_q|}\sum_{k=1}^{n_q} P_q(k)rel_q(k) \tag{7}$$

where $P_q(k)$ is the precision at position k for query q.

The effectiveness with which a system returns the first relevant result is measured by the Mean Reciprocal Rank (MRR), defined as:

$$\text{MRR} = \frac{1}{|Q|}\sum_{q\in Q}^{K}\frac{1}{rank_q} \tag{8}$$

where $rank_q$ is the position of the first relevant fragment.

Finally, Top $-$ 1Accuracy, a special case of Precision@K with $K = 1$, indicates the percentage of queries for which the first result is relevant:

$$\text{Top} - 1\text{Accuracy} = \frac{1}{|Q|}\sum_{q\in Q} rel_q(1) \tag{9}$$

The combination of these metrics offers a comprehensive view of retriever performance, from immediate relevance (Top $-$ 1Accuracy) to overall coverage (Recall@K) and ranking quality (MAP and MRR) [58], [59].

ii) Text-to-SQL agent

The evaluation of an agent specializing in public procurement was based on two metrics widely recognized in the literature: Execution Accuracy (EX) and False-Less Execution (FLEX), originally proposed by Yu et al. in the Spider benchmark [60] and later extended by Kim et al. as a semantic evaluation approach using large language models [61].

The first metric, Execution Accuracy (EX), measures the proportion of automatically generated queries that, when executed on the database, produce exactly the same results as the reference queries. This metric is formalized as:

where N is the total number of evaluated questions, q_i^{gold} is the query generated by the system, q_i^{gold} is the reference query $Exec\,(.)$ represents the query execution, and $1\,(.)$ is the indicator function that returns to 1 if both executions are identical.

$$\text{EX} = \frac{1}{N}\sum_{i=1}^{N} 1\left(Exec\left(q_i^{pred}\right) = Exec\left(q_i^{gold}\right)\right) \tag{10}$$

However, in many real-world cases, multiple valid SQL queries may correspond to the same natural language question. To capture this variability and better reflect the semantic intent of the user, the FLEX metric was employed. FLEX is a recent metric that delegates correctness judgments to an advanced language model. In this approach, each generated query is evaluated via prompts designed to allow an LLM to determine whether it adequately answers the original question regardless of textual similarity to a reference. FLEX is defined as:

$$\text{FLEX} = \frac{1}{N}\sum_{i=1}^{N} f_i \tag{11}$$

where f_i is a binary verdict provided by the model (1 if the query is semantically valid, and 0 otherwise). This metric has demonstrated a significantly higher agreement with human evaluators and has become the new standard for Text-to-SQL tasks in contexts where semantic interpretation is a priority.

4) RESULTS

i)Performance of RAG-Based Agents

RAG-based agents, developed for congressional attendance and voting queries, demonstrated robust when compared against established baselines. A benchmark of 200 manually curated queries—100 for attendance and 100 for voting—was constructed from (i) the structured

TABLE 11. Performance metrics of RAG agents for attendance.

Metric	Qdrant	BM25	TF-IDF
Precision @10	0.930 [0.878, 0.973]	0.465 [0.404, 0.527]	0.101 [0.061, 0.145]
Recall @10	0.964 [0.924, 0.994]	0.817 [0.740, 0.890]	0.244 [0.163, 0.330]
Top-1 Accuracy	0.940 [0.890, 0.980]	0.761 [0.670, 0.840]	0.130 [0.070, 0.200]
Average Rank	1.104 [1.000, 1.263]	1.263 [1.070, 1.516]	3.757 [2.483, 5.053]
MAP	0.950 [0.906,0.986]	0.786 [0.706, 0.862]	0.165 [0.100, 0.235]
MRR	0.949 [0.904, 0.985]	0.783 [0.702, 0.860]	0.151 [0.088, 0.220]

datasets presented in previous sections (attendance records; voting headers, subjects, and session presidents), and (ii) query patterns inspired by investigative journalism on congressional transparency, including Convoca [62], El Comercio [63] and Ojo Público [64]. This benchmark ensured both grounding in real data and alignment with practical information needs. Results should be interpreted as context-specific to targeted query types; broader or less structured queries may yield lower retrieval performance.

Semantic retrieval with Qdrant was compared against two widely used lexical baselines: BM25, a probabilistic relevance model [65], and TF-IDF, a term-weighting scheme frequently applied in query expansion and retrieval tasks [66].

For statistical rigor, mean values were reported with 95% confidence intervals estimated through percentile bootstrap resampling with 2,000 iterations, balancing computational cost with interval precision. Bootstrap confidence intervals are widely recommended for non-normally distributed evaluation metrics in NLP and information retrieval tasks, including ranking and semantic similarity measures [32], [33], [53].

As shown in Table 11 (attendance), Qdrant achieved near-perfect performance: Precision@10 = 0.930 [0.878, 0.973], Recall@10 = 0.964 [0.924, 0.994], Top-1 Accuracy = 0.940 [0.890, 0.980], MAP = 0.950 [0.906, 0.986], and MRR = 0.949 [0.904, 0.985], with average ranks close to the first position. Both BM25 and TF-IDF scored significantly lower across all metrics.

In Table 12 (voting), Qdrant again outperformed the baselines, reaching Precision@10 = 0.791 [0.721, 0.856], Recall@10 = 0.855 [0.801, 0.903], and Top-1 Accuracy = 0.900 [0.840, 0.950], while MAP and MRR reached 0.926 [0.878, 0.967]. Performance was slightly lower than in attendance queries, primarily due to higher noise and ambiguity in questions about the session president field, which introduced variability in retrieval. Nevertheless, semantic retrieval via Qdrant consistently and significantly surpassed BM25 and TF-IDF.

Additionally, a stress set of 50 queries was derived from manually selected attendance and voting documents with

TABLE 12. Performance metrics of RAG agents for voting.

Metric	Qdrant	BM25	TF-IDF
Precision@ 10	0.791 [0.721, 0.856]	0.154 [0.123, 0.188]	0.040 [0.028, 0.052]
Recall @10	0.855 [0.801, 0.903]	0.498 [0.413, 0.582]	0.256 [0.174, 0.340]
Top-1 Accuracy	0.900 [0.840, 0.950]	0.370 [0.270, 0.470]	0.221 [0.140, 0.300]
Average Rank	1.145 [1.041, 1.277]	2.778 [2.228, 3.377]	2.485 [1.656, 3.407]
MAP	0.926 [0.878, 0.967]	0.477 [0.396, 0.558]	0.248 [0.170, 0.330]
MRR	0.926 [0.878, 0.967]	0.483 [0.398, 0.568]	0.247 [0.169, 0.329]

TABLE 13. Retrieval Performance Under OCR Stress Set.

Metric	Qdrant	BM25	TF-IDF
Precision@ 10	0.680 [0.610, 0.744]	0.126 [0.092, 0.164]	0.035 [0.022, 0.049]
Recall @10	0.752 [0.680, 0.816]	0.413 [0.330, 0.495]	0.230 [0.150, 0.314]
Top-1 Accuracy	0.765 [0.660, 0.840]	0.296 [0.195, 0.390]	0.179 [0.100, 0.255]
Average Rank	1.317 [1.120, 1.560]	3.112 [2.500, 3.920]	2.834 [1.800, 3.950]
MAP	0.806 [0.730, 0.870]	0.401 [0.315, 0.485]	0.211 [0.140, 0.286]
MRR	0.808 [0.735, 0.872]	0.406 [0.320, 0.490]	0.210 [0.140, 0.286]

the poorest quality and the highest visual noise caused by overlapping stamps, corresponding to the worst-performing regions in the OCR evaluation reported in Table 10. The evaluation, using 2,000 percentile bootstrap resamples, showed that Qdrant outperformed BM25 and TF-IDF under adverse OCR conditions (Table 13).

These findings confirm that RAG-based agents provide accurate and reliable retrieval in both attendance and voting scenarios, with statistically validated improvements over lexical methods.

ii) Performance of Text-to-SQL Agent

The public procurement agent was evaluated using a benchmark of 120 manually curated queries. These queries were inspired by investigative journalism on government procurement, including reports from La República [67] and IDL Reporteros [68], as well as by the methodological aspects highlighted in the Funes framework proposed by Ojo Público [69]. The benchmark was not stratified by levels; instead, it covered a diverse set of queries with varying degrees of complexity, ranging from direct lookups and aggregations to more advanced analytical tasks such as contract periodicity, contract-type distributions, and filtering

TABLE 14. Representative examples of queries in the benchmark.

Complexity Level	Example Query (English translation)	Key Operations Required
Simple	What is the name of the supplier with RUC 20100117526?	Single-table lookup with direct filtering (no joins, no aggregation).
Complex	How many suppliers that began operations from February 2020 onward have signed service orders with the Municipalidad Distrital de Los Olivos? Return the top 5 ranked by number of service orders and the total contracted amount.	Multi-table join + date comparison + semantic fuzzy matching in entity names + grouped aggregation + descending ordering + limit.

contracts signed after the supplier's registration date (see Table 14 for representative examples).

To ensure statistical rigor, results were reported using EX and FLEX, together with 95% confidence intervals estimated via percentile bootstrap resampling with 2,000 iterations, a widely recommended approach for evaluation metrics that do not follow a normal distribution [32], [33], [53].

Three approaches were compared. The first was a General Script, a non-reasoning baseline aligned with early rule- and template-based methods in Natural Language Interfaces to Databases (NLIDB) [70], [71], noting that template-based baselines have also been employed in dialog-oriented components such as response generation [72]. In practical terms, this baseline uses basic keyword matching to choose the most similar predefined SQL template and then fills in its variable fields, without any semantic interpretation or reasoning. The second was a Simple Agent, implemented as a single-node GPT-4o model with a prompt explicitly including the tables and the database schema, consistent with recent LLM-based Text-to-SQL approaches [73]. The third was Our Agent, which integrates schema-aware prompting (aligned with schema linking practices), semantic validation, retry mechanisms upon execution failure, and direct access to database functions, and is designed to adapt dynamically to schema changes [71], [74].

The results, summarized in Table 15, demonstrate that our agent consistently outperformed both baselines. Across the entire benchmark, it achieved the highest FLEX and EX scores, maintaining robust accuracy even as query complexity increased. Overall, our agent reached FLEX = 0.909 [0.851, 0.959] and EX = 0.801 [0.727, 0.868], compared to FLEX = 0.798 [0.727, 0.868] and EX = 0.700 [0.619, 0.785] for the simple agent, and FLEX = 0.685 [0.595, 0.769] and EX = 0.660 [0.578, 0.743] for the general script.

C. ARTIFACT 3: MULTI-AGENT ORCHESTRATED SYSTEM

1) PURPOSE

The third artifact addresses the critical challenge of integration and accessibility that persists after developing Artifacts 1 and 2. While these artifacts transform heterogeneous governmental data and enable specialized retrieval, their outputs remain technical and fragmented, limiting their value for citizen transparency.

TABLE 15. Evaluation of the Text-to-SQL agent using EX and FLEX.

Agent	EX	FLEX
General Script	0.660 [0.578, 0.743]	0.685 [0.595, 0.769]
Simple Agent	0.700 [0.619, 0.785]	0.798 [0.727, 0.868]
Our Agent	0.801 [0.727, 0.868]	0.909 [0.851, 0.959]

The purpose of this artifact is to provide a unified orchestration system that:

- Coordinates specialized agents into a cohesive workflow.
- Exposes a single access point that reduces service fragmentation.
- Enables natural-language queries without requiring technical knowledge.
- Synthesizes outputs into contextualized and auditable responses.
- Ensures scalability through the web platform.

In this way, the artifact extends the contributions of the previous components by transforming technical retrieval processes into a citizen-facing service that directly supports governmental transparency.

2) DESIGN

The design of the multi-agent orchestration system specifies how integration is achieved through Service-Oriented Architecture (SOA) principles and the microservice paradigm [40]. Lercher et al. [40] highlight that microservices improve scalability, maintainability, and fault tolerance by relying on loose coupling and independent domain models, while Roca et al. [39] demonstrate their advantages for conversational systems in terms of flexibility and extensibility. Building on these foundations, in our system microservices are further leveraged to provide modularity, enabling agents and pipelines to evolve independently while supporting concurrent queries through the web interface. Within this context, the orchestration layer becomes the focal point of the design, as it coordinates specialized agents [18], [19], manages user queries, and integrates outputs into contextualized and auditable responses. To ensure reproducibility and clarity, the architecture is documented using the C4 model [75], and the following subsections detail the overall architecture, orchestrated components, conversational flow, architectural blueprint, and design considerations relevant to conversational evaluation.

a: OVERALL ARCHITECTURE

The overall architecture is described using the C4 model at the container level, presenting the main services and their

TABLE 16. Evaluation of the conversational system.

Metric	Attendance	Voting	Procurement	All
Faithfulness	0.957 [0.932, 0.977]	0.977 [0.955, 0.994]	0.910 [0.894, 0.925]	0.942 [0.930, 0.954]
Answer Relevancy	0.907 [0.867, 0.940]	0.937 [0.904, 0.955]	0.890 [0.865, 0.908]	0.907 [0.889, 0.923]
Answer Similarity	0.987 [0.983, 0.990]	0.975 [0.972, 0.977]	0.986 [0.983, 0.988]	0.984 [0.982, 0.985]
Answer Correctness	0.905 [0.867, 0.939]	0.858 [0.800, 0.907]	0.899 [0.874, 0.922]	0.891 [0.871, 0.911]
BERTScore F1	0.773 [0.727, 0.812]	0.761 [0.737, 0.781]	0.764 [0.733, 0.795]	0.762 [0.741, 0.783]
BLEU	0.679 [0.633, 0.721]	0.602 [0.564, 0.640]	0.634 [0.588, 0.679]	0.642 [0.617, 0.667]
ROUGE-1 F1	0.800 [0.766, 0.829]	0.831 [0.812, 0.850]	0.813 [0.788, 0.837]	0.813 [0.797, 0.828]
ROUGE-2 F1	0.692 [0.655, 0.725]	0.734 [0.713, 0.754]	0.701 [0.661, 0.738]	0.706 [0.686, 0.726]
ROUGE-L F1	0.762 [0.726, 0.796]	0.607 [0.587, 0.629]	0.748 [0.713, 0.783]	0.719 [0.698, 0.738]

interactions. The web application acts as the entry point with the conversational interface for end users. Queries are routed to the AI orchestrator, which manages the workflow and invokes specialized agents according to the query domain. These agents include two RAG agents for attendance and voting, and one SQL agent for procurement. They interact with the relational and vector databases as required, and the LLM API supports query understanding and response synthesis. For clarity, the document service is not shown, as Figure 10 emphasizes the core conversational workflow.

b: ORCHESTRATED COMPONENTS

The integration is structured around the following core components:

- **Gateway service**: unified entry point exposing REST APIs and WebSocket endpoints, handling authentication and normalizing requests from web clients.
- **Chats Service**: orchestration service governing the conversational workflow. Implemented with LangGraph, it dynamically invokes MCP agents through well-defined contracts and synthesizes outputs into a uniform response format.
- **Documents Service**: complementary service enabling direct queries (by company ID or RUC) without requiring conversational interaction.
- **MCP agents**: attendance, voting, and procurement agents. These encapsulate the domain-specific logic developed in previous artifacts and expose standardized interfaces for seamless orchestration.
- **Persistence layer**: Qdrant for semantic vector retrieval and PostgreSQL databases following a database-per-service pattern, which facilitates scaling and fault isolation.

c: CONVERSATIONAL FLOW

The orchestration of conversational interactions is implemented as a workflow in LangGraph, modeled as a directed

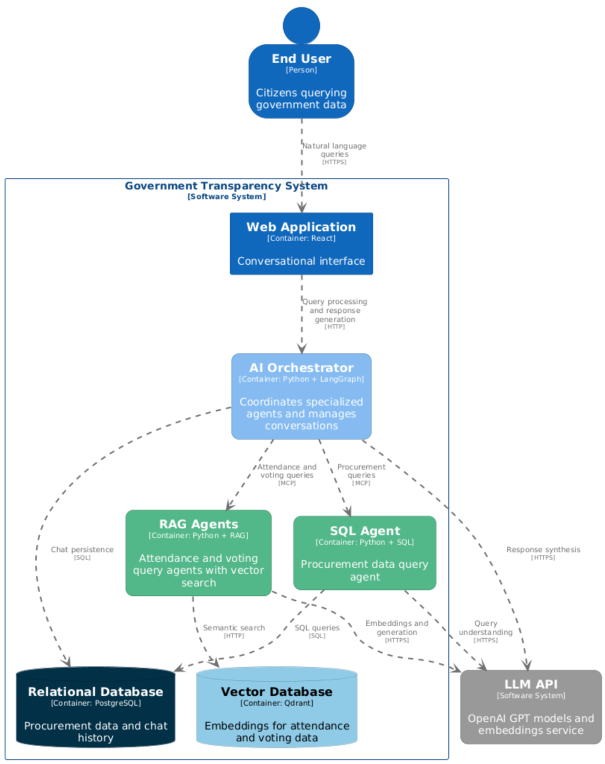

FIGURE 10. C4 Container diagram of the proposed system.

graph of nodes and tools. As shown in Figure 11, the process begins at the classifier node, which determines whether the user query is in-domain or out-of-domain. Out-of-domain queries are redirected to the fallback node, which generates a safe and consistent default response before reaching the end node. Valid queries proceed to the main node, responsible for processing the input and invoking the appropriate tools, corresponding to the MCP agents. The results are returned to the main node, consolidated, and then passed to the end node as a standardized response schema, ensuring both consistency and traceability. Furthermore, Figure 12 illustrates an example of a real conversation in the web interface, demonstrating how the orchestration is reflected in practice.

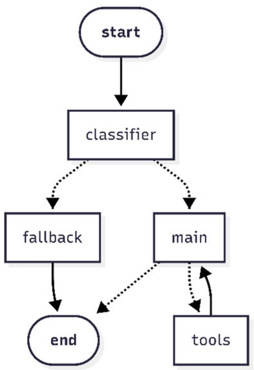

FIGURE 11. Conversational workflow implemented in LangGraph.

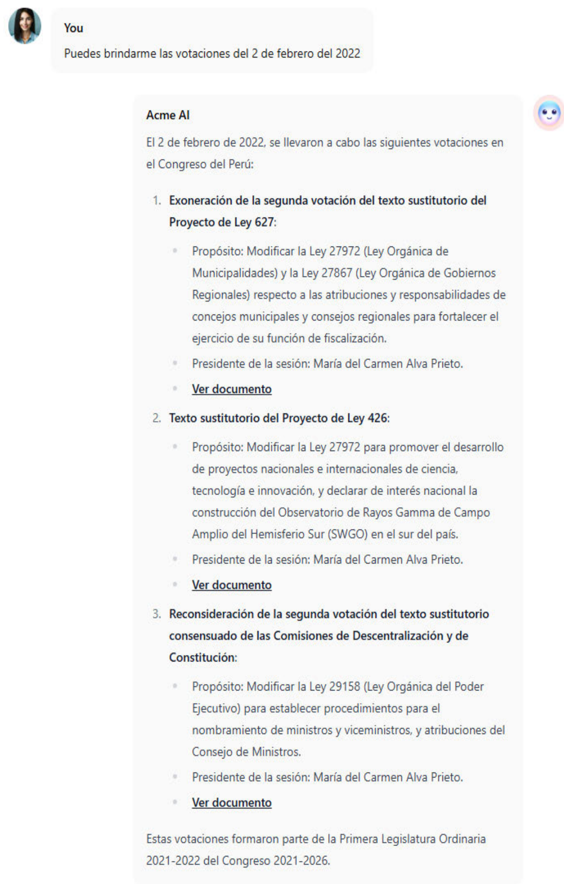

FIGURE 12. Web interface showing an example query and response about parliamentary voting.

d: ARCHITECTURAL BLUEPRINT

The architectural blueprint corresponds to the C4 component view, complementing the container-level description presented earlier. As shown in Figure 13, this view specifies the system at the component level, providing a detailed and reproducible blueprint that extends the overall architecture into a concrete specification.

3) METRICS

In Artifact 2, the evaluation addressed retrieval performance for the RAG-based agents and execution accuracy for the Text-to-SQL agent, ensuring that the system could provide the necessary contextual information.

In this artifact, the focus shifts to assessing the end-to-end generation quality, where all agents must produce responses that are factually accurate, semantically relevant, consistent with the retrieved evidence, and linguistically fluent.

To this end, two complementary groups of metrics were employed: the generation-oriented components of the RAGAS framework and classical NLG metrics, both validated in indexed literature.

The RAGAS framework, originally designed to assess Retrieval-Augmented Generation systems in open environments [30], [31], is here applied exclusively to its generation-focused metrics, which are described in detail below.

Faithfulness measures whether the claims made in the generated answer are supported by the retrieved context. The system decomposes the answer into a set of statements S (as (q)), and each statement is verified against the context c (q). Let $V \subseteq S$ be the subset of supported statements. The final score is computed as:

$$\text{Faithfulness} = \frac{|V|}{|S|} \tag{12}$$

where $|V|$ is the number of supported statements and $|S|$ is the total number of statements in the answer.

Answer Relevancy (AnsRel) evaluates to what extent the generated answer directly addresses the user's question. For a given answer, n synthetic questions q_i are generated and compared to the original question q using embeddings. The score is defined as:

$$\text{AnsRel} = \frac{1}{n}\sum_{\text{i}=1}^{\text{n}} \cos\left(\text{E}\left(q\right), \text{E}\left(q_i\right)\right) \tag{13}$$

where $\text{E}(q)$ and $\text{E}(q_i)$ denote the embeddings of the original and synthetic questions, respectively.

Answer Similarity (AnsSim) measures the semantic closeness between the generated response $a(q)$ and the ground truth answer $gt(q)$. It is computed as the cosine similarity of their embeddings:

$$\text{AnsSim} = \cos\left(\text{E}\left(a(q)\right), \text{E}\left(gt(q)\right)\right) \tag{14}$$

where $\text{E}(\cdot)$ denotes the embedding function.

Answer Correctness (AnsCor) evaluates the overall correctness of the generated answer with respect to the ground

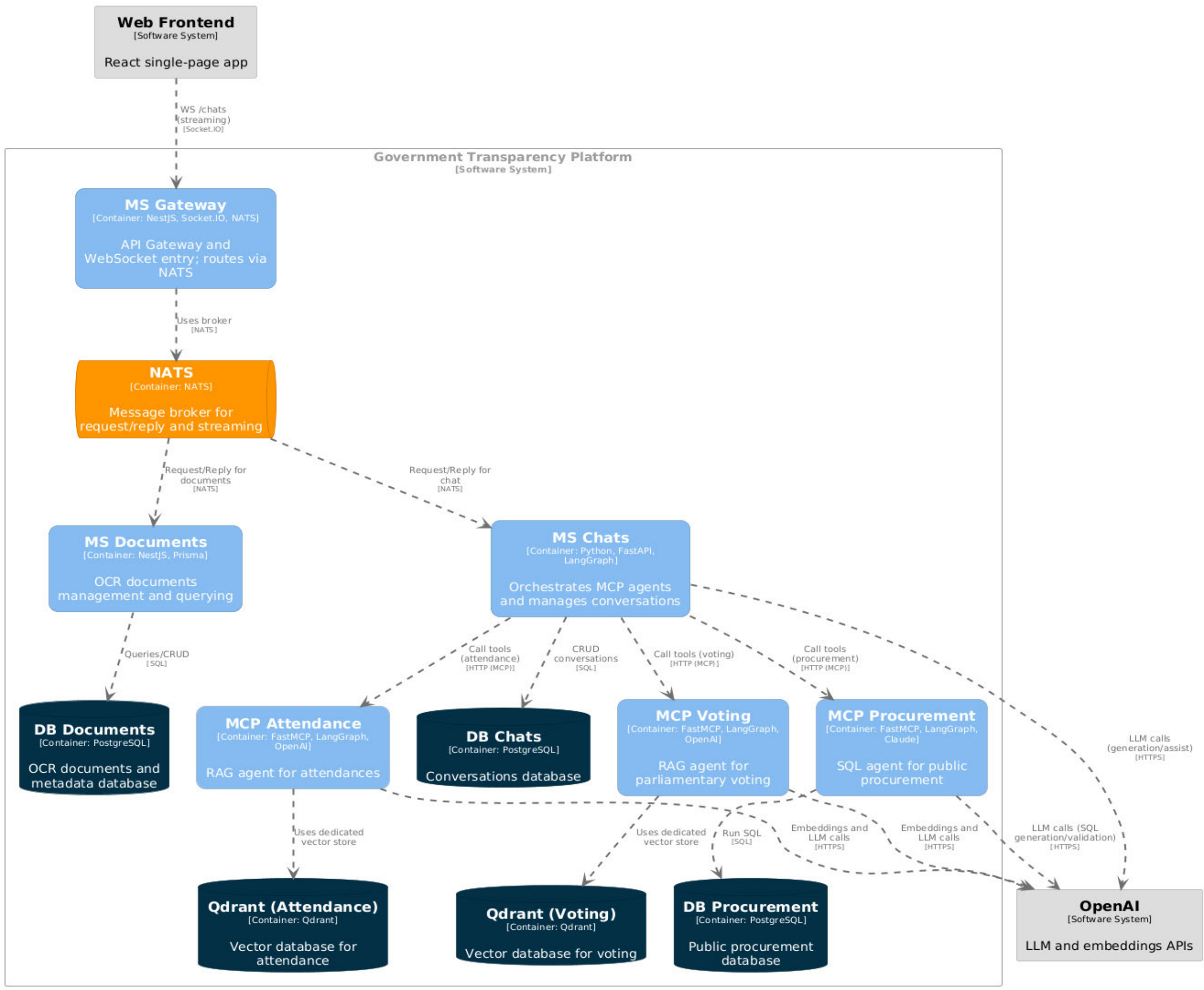

FIGURE 13. C4 component diagram of the proposed system.

truth by combining factual correctness and semantic similarity. Let TP, FP, and FN be the sets of true positive, false positive, and false negative statements, respectively. Factual correctness (FacCor) is defined as:

$$\mathrm{FacCor} = \frac{|\mathrm{TP}|}{|\mathrm{TP}| + 0.5 \times (|\mathrm{FP}| + |\mathrm{FN}|)} \tag{15}$$

and the final score is given by a weighted combination of factual correctness and answer similarity:

$$\mathrm{AnsCor} = \omega_1 \times \mathrm{FacCor} + \omega_2 \times \mathrm{AnsSim} \tag{16}$$

with default weights $\omega_1 = 0.75$ and $\omega_2 = 0.25$.

On the NLG metrics, the evaluation incorporated a set of classical measures that have been widely validated in indexed literature. These metrics, BERT Score, BLEU and ROUGE, focus on aspects such as semantic similarity, lexical overlap and linguistic fluency, thereby complementing the RAGAS measures and providing a broader view of generation quality, as described in the following.

$\mathrm{BERTScore}_{\mathrm{F1}}$ computes semantic similarity using contextual embeddings from transformer models, aligning tokens between candidate and reference texts. It is defined as:

$$\mathrm{BERTScore}_{\mathrm{F1}} = \frac{2 \cdot P \cdot R}{P + R} \tag{17}$$

where P and R denote precision and recall over embedding-based matches [76].

BLEU (Bilingual Evaluation Understudy) measures the precision of n-gram matches between candidate and reference texts, with a brevity penalty (BP) to avoid favoring short outputs:

$$\mathrm{BP} = \begin{cases} 1, \ell_c > & \ell_r \\ e^{1-\ell_r/\ell_c}, & \ell_c \leq \ell_r \end{cases} \tag{18}$$

$$\text{BLEU} = BP \cdot exp\left(\sum_{n=1}^{N} \omega_N \log p_n\right) \tag{19}$$

where p_n is the modified precision of n-grams, ω_n their weights, and ℓ_c, ℓ_r the candidate and reference lengths [77], [78].

ROUGE evaluates overlap between generated and reference texts. ROUGE-1 considers unigrams, ROUGE-2 bigrams, and ROUGE-L the longest common subsequence. Its F1 form is expressed as:

$$\text{ROUGE} - \text{NF1} = \frac{2 \cdot P_N \cdot R_N}{P_N + R_N} \tag{20}$$

where P_N and R_N correspond to precision and recall of n-grams of order N [79], [80].

4) RESULTS

The overall performance of the conversational system was evaluated over a benchmark of 300 queries: 100 on congressional attendance, 100 on voting, and 100 on public procurement. The evaluation combined the RAGAS framework with classical NLG metrics (BERTScore, BLEU, and ROUGE), enabling a comprehensive assessment of both semantic fidelity and linguistic quality.

To ensure statistical rigor, all RAGAS and NLG metrics were reported with 95% confidence intervals estimated via percentile bootstrap resampling with 2,000 iterations. As highlighted in [53], bootstrap intervals are preferable for NLP evaluation because metric distributions often deviate from normality, making classical parametric assumptions unreliable. In [32], the authors recommend bootstrap resampling as part of best practices in Natural Language Generation evaluation, stressing the importance of reporting confidence intervals alongside metric scores. Finally, [33] demonstrates that bootstrap provides stable and interpretable intervals for semantic evaluation metrics, reinforcing its suitability for tasks involving ranking and similarity measures.

The results demonstrated strong performance across all metrics. The system achieved high scores in Faithfulness (0.942), Answer Similarity (0.984), and Answer Relevancy (0.907), while classical NLG metrics showed consistent outcomes: BERTScore F1 (0.762), BLEU (0.642), and ROUGE-L F1 (0.719). These findings confirm the robustness of the model in both semantic fidelity and linguistic quality (Table 16).

V. DISCUSSIONS

A. SYSTEM PERFORMANCE AND TECHNICAL CONTRIBUTIONS

The results demonstrate the technical feasibility of integrating multiple artificial intelligence technologies to address the challenges of government transparency in the context of poor data quality. The proposed architecture achieved outstanding performance across its main components. In the classification stage, deep learning models such as ResNet50, VGG16, and DenseNet121 achieved cross-validated accuracies above 99% with test accuracies up to 0.989, confirming their reliability for distinguishing relevant legislative documents (Table 7). In the region detection task, YOLOv11 models obtained robust results, with the best attendance configuration reaching mAP@50–95 = 0.965 [0.962, 0.969] and the best voting configuration reaching 0.886 [0.873, 0.899] (Tables 8–9). In the OCR stage, Tesseract consistently outperformed alternatives, achieving an average CER of 0.031 [0.020, 0.042] and WER of 0.078 [0.060, 0.098] (Table 10), an order of magnitude lower than EasyOCR or PaddleOCR in key header regions. These results are competitive with those reported in previous studies, such as Liu et al. [25], who achieved a 52% improvement in recall compared to traditional methods.

The combination of RAG and Text-to-SQL agents also represents a significant contribution. The attendance agent achieved near-perfect retrieval performance with Precision@10 = 0.930 [0.878, 0.973], Recall@10 = 0.964 [0.924, 0.994], MAP = 0.950 [0.906, 0.986], and MRR = 0.949 [0.904, 0.985] (Table 11). The voting agent also demonstrated strong performance with MAP and MRR = 0.926 [0.878, 0.967] and Top-1 Accuracy of 0.900 [0.840, 0.950] (Table 12).

These results align with evidence from specialized systems in the medical field, where the integration of RAG improved accuracy from 43% to 99% [16], confirming the robustness of semantic retrieval beyond domain-specific boundaries. Importantly, even under adverse conditions characterized by poor scan quality and overlapping stamps—the OCR Stress Set—Qdrant maintained Precision@10 = 0.680 [0.610, 0.744], MAP = 0.806 [0.730, 0.870], and MRR = 0.808 [0.735, 0.872], clearly outperforming BM25 and TF-IDF by large margins (Table 13). This robustness highlights the system's capacity to preserve retrieval quality even in challenging real-world scenarios.

For structured procurement queries, the Text-to-SQL agent outperformed both baseline approaches. It reached FLEX = 0.909 [0.851, 0.959] and EX = 0.801 [0.727, 0.868], compared to FLEX = 0.798 and EX = 0.700 for the simple agent, and FLEX = 0.685 and EX = 0.660 for the general script (Table 15). These results validate the agent's ability to generate semantically correct and executable queries, confirming the effectiveness of the FLEX metric introduced by Kim et al. [61] as a benchmark for semantic correctness in real-world Text-to-SQL tasks.

The integrated conversational system achieved solid end-to-end performance. Across 300 benchmark queries, the system obtained Faithfulness = 0.942 [0.930, 0.954], Answer Relevancy = 0.907 [0.889–0.923], Answer Similarity = 0.984 [0.982, 0.985], and Answer Correctness = 0.891 [0.871, 0.911]. Complementary NLG metrics also showed robust linguistic quality, with BERTScore F1 = 0.762, BLEU = 0.642, and ROUGE-L F1 = 0.719 (Table 16).

Finally, although BLEU and ROUGE scores were comparatively lower (0.642, 0.719), this outcome is consistent with the lexical variability. These surface-form metrics penalize valid paraphrases and alternative formulations, which are common in legislative and procurement data. In contrast, semantic-oriented measures such as BERTScore and RAGAS confirmed that the generated answers remained factually accurate, contextually relevant, and aligned with user intent. This reinforces the robustness of the proposed multi-agent architecture and the importance of combining classical and semantic evaluation metrics when addressing the complexity of governmental transparency tasks.

B. IMPLICATIONS FOR GOVERNMENT TRANSPARENCY

The findings of this study have direct implications for government transparency in Latin America. While countries such as Mexico, Uruguay, and Brazil lead in open data policies [4], and others like Argentina have experienced regulatory setbacks [5], the proposed system offers a technological alternative to overcome the structural limitations identified in the Peruvian context.

The developed pipeline for transforming low-quality scanned documents directly addresses the limitations imposed by the persistence of unstructured and low-resolution formats in the public sector [13], [14]. By demonstrating how to convert these "Data Graveyards" [15] into structured and searchable information through classification, region detection, and OCR techniques, the system establishes a methodology capable of compensating for deficiencies in governmental data policies.

The microservices architecture facilitates the scalability required to address the lack of open data portals at the subnational level, where only 0.26% of the regional and local governments have adequate platforms [14]. This modularity enables gradual implementation and jurisdiction-specific adaptations.

C. COMPARATIVE ANALYSIS WITH INTERNATIONAL EXPERIENCES

International experiences reveal uneven progress in government automation and open data. China and the United States stand out for scale and public sector receptivity [17], [24], while Latin America shows more heterogeneous outcomes: civic platforms such as Por Mi Barrio in Uruguay highlight the potential of NGO–government collaboration [7]; in Brazil, significant gaps remain in state-level open data use [8] and in transparency during the pandemic [9]; and across 18 countries, sustainability reporting is still constrained by systemic factors such as legislation quality and corruption [10].

Against this fragmented landscape, our model offers three distinctive contributions: (1) integration of multiple governmental domains into a single interface; (2) the ability to process heterogeneous documents without prior standardization; and (3) a conversational architecture that lowers technical barriers and broadens citizens' oversight capacity.

D. LIMITATIONS AND FUTURE WORK

Despite the promising results, the system presents limitations that warrant consideration. The reliance on commercial language models introduces operational costs and data sovereignty concerns that may affect long-term sustainability. To address these constraints, future work should evaluate the adoption of optimized open-source models adapted to both international and Peruvian governmental contexts. Building on this direction, LLMs could serve as higher-level refinement or orchestration layers that validate SQL queries, clarify ambiguous questions, synthesize information across retrieved documents, and produce more accurate natural-language explanations. This hybrid integration would enhance the system's expressiveness and robustness while preserving its transparency, modularity, and cost-efficiency.

The temporal coverage of the system (2006–2025) reflects the historical availability of digitized records from the Peruvian Congress, with 2006 as the earliest year of systematic documentation. A key limitation of the current implementation is the absence of automated pipelines for incorporating newly generated governmental data. Continuous ingestion mechanisms were outside the scope of this study; therefore, updates beyond the cut-off date were not implemented. Nevertheless, the project repository includes a *publicdata-scraper* module that provided the foundation for the initial data collection. This component constitutes a concrete starting point for developing automated or periodic update pipelines in future iterations, ensuring that the system remains aligned with newly published procurement records and congressional documents. Future work will extend this module into a robust ingestion framework with scheduled retraining and version control mechanisms to ensure long-term sustainability.

The current monolingual design of the system represents an additional limitation, as it only supports queries and documents in Spanish. This restricts accessibility for indigenous language speakers, particularly Quechua and Aymara communities, who represent a significant share of the Peruvian population. While multilingual support was beyond the scope of this study, the modular architecture allows extensions with additional language models and parallel corpora. Future work will focus on integrating multilingual embeddings and machine translation pipelines to enable equitable access across linguistic groups.

Additionally, since the system is capable of processing both structured and semi-structured data, future work should expand its coverage to include additional public repositories. One key source is the Official Record of Debates of the Peruvian Congress, which provides the official transcripts of all plenary sessions. Other valuable additions include records from permanent commissions, committee reports, and official gazettes such as the Official Gazette of Peru (El Peruano). Incorporating these sources would significantly enhance the system's comprehensiveness, ensuring that citizens gain access to a richer and more diverse body of governmental information through a unified conversational interface.

E. ETHICAL AND GOVERNANCE CONSIDERATIONS

As the proposed system manages sensitive public information to enhance government transparency, it requires ethical reflection and an adequate governance framework for responsible implementation. The main risk is the misuse or misinterpretation of generated responses, particularly in political or judicial contexts. To mitigate this risk, the system was designed under the principle of factual faithfulness, restricting the generation to retrieved documentary fragments or traceable SQL queries. In the case of attendance and voting records in Congress, official supporting documents are shown directly, which reinforces the legitimacy of each response and enables user verification.

It is also recommended that any large-scale implementation should be accompanied by an algorithmic governance framework that includes institutional oversight, citizen participation, and transparency in the model's operation. Although this thesis works exclusively with public data, future system extensions may require additional mechanisms, such as anonymization, access control, or external audits, particularly if new data sources are integrated.

This approach aligns with the OECD guidelines on the responsible use of artificial intelligence in the public sector and aims to prevent both the misuse of these technologies and public mistrust in their adoption [81].

VI. CONCLUSION

This study presents a multi-agent conversational system designed to improve government transparency in Peru by intelligently integrating heterogeneous public information through language interfaces. The developed hybrid architecture combines advanced document processing with specialized RAG and Text-to-SQL agents coordinated via the Model Context Protocol, transforming over 16,000 legislative documents and 12 million procurement records into queryable informational resources.

The experimental results confirmed the system's effectiveness: document classification with test accuracy up to 0.9889, region detection with mAP@50–95 up to 0.965 for attendance and 0.886 for voting, OCR error rates with Tesseract achieving an average CER of 0.0305 and WER of 0.0777, RAG agents with MAP = 0.950 (attendance) and 0.926 (voting), and the Text-to-SQL agent with FLEX = 0.909. At the conversational level, the system achieved Faithfulness = 0.942, Answer Relevancy = 0.907, Answer Correctness = 0.891, and Answer Similarity = 0.984. Complementary natural language generation metrics also demonstrated high-quality outputs, with BERTScore F1 = 0.762, BLEU = 0.642, and ROUGE-L F1 = 0.719, confirming that responses are not only factually accurate but also linguistically coherent and well-structured. These combined metrics demonstrate the system's ability to generate contextually accurate and factually grounded responses to complex queries regarding government information, even under adverse OCR conditions.

Beyond technical achievements, the fundamental contribution of this study lies in establishing a replicable methodology that shows how artificial intelligence technologies can overcome the structural limitations that hinder government transparency in resource-constrained environments. The developed pipeline for processing low-quality scanned documents offers a practical solution to the systematic shortcomings of government repositories, whereas the modular microservice architecture enables gradual deployment and jurisdiction-specific adaptations.

This methodology transcends the Peruvian context, offering a framework applicable to other Latin American countries facing the similar challenges of informational fragmentation and low-quality government data. By demonstrating the technical feasibility of transforming static governmental repositories into intelligent conversational systems, this study makes a significant contribution to strengthening accountability and enabling citizen access to critical public information.

The results obtained lay the groundwork for a paradigm shift in the application of artificial intelligence to government transparency. The transition toward unified conversational systems represents a fundamental advancement that can strengthen democratic governance through equitable and efficient access to public information, fostering more informed and effective citizen participation.

IMPLEMENTATION AND INFRASTRUCTURE

The system was deployed using a microservices architecture with Docker Compose. Training and preprocessing were performed on a local workstation, while evaluation and benchmarking were carried out on an AWS EC2 c5.4xlarge instance (16 vCPUs, 32 GiB RAM). Table 17 summarizes the hardware resources for both environments, including the storage footprint of the main subsystems.

TABLE 17. Hardware configuration and deployment environments.

Component	Training Environment	Testing/Production Environment
CPU	AMD Ryzen 7 7800X3D	AWS EC2 c5.4xlarge (16 vCPUs)
GPU	NVIDIA RTX 4070 Ti (16 GB)	N/A
RAM	64 GB DDR5	32 GiB
Storage	2 TB NVMe SSD	100 GB EBS gp3

Latency values were obtained from AWS CloudWatch logs during system execution. Table 18 reports average and 95th-percentile (p95) latencies at the subsystem level and for end-to-end queries.

Operational costs were estimated for 30,000 queries per month, combining infrastructure and API usage.

TABLE 18. Latency metrics.

Task	Mean Latency (ms)	p95 (ms)
PostgreSQL query	78	115
Qdrant search	48	60
Nginx static file	19	22
RAG end-to-end	2,125	2,350
Text-to-SQL end-to-end	1,495	1,700

TABLE 19. Operational costs.

Category	Cost per Query (USD)	Monthly Cost (USD) GPT-4o
RAG query	0.0120	360.0
Text-to-SQL query	0.0252	756.0
Mixed workload (50% RAG / 50% T-SQL)	—	558.0
Infrastructure (EC2 + EBS)	—	504.4
Total (infra + API, mixed workload)	—	1,062.2

TABLE 20. Summary of repository modules and functions.

Submodule	Domain / Function
frontend/	Web application interface
ms-gateway/	API gateway; REST & WebSocket communication
ms-chats/	Conversational microservice; MCP orchestration
ms-documents/	Document retrieval microservice
mcp-attendance/	Attendance agent (RAG with Qdrant)
mcp-voting/	Voting agent (RAG with Qdrant)
mcp-procurement/	Procurement agent (dynamic Text-to-SQL)
publicdata-scraper/	Data extraction from OSCE, SUNAT and Congress Portal.
publicdata-classifier/	Dataset preparation, training, and evaluation for document classification
publicdata-yolo-ocr/	Datasets, training, and evaluation of YOLO models and OCR modules
train-main-llm/	Cross-domain datasets and evaluation scripts
train-voting/	Datasets, evaluation, and Qdrant indexing for voting agent
train-procurement/	Datasets, evaluation, and PostgreSQL loading for procurement agent
train-attendance/	Datasets, evaluation, and Qdrant indexing for attendance agent
docker-compose.yml	Root-level orchestration file for deploying the entire project

Infrastructure costs correspond to EC2 compute and EBS storage, while API costs vary depending on the selected model. Table 19 summarizes these results.

Subsystem-level operations such as database queries, vector search, and static file serving consistently remained below 100 ms (p95), whereas end-to-end queries including LLM inference required 1.5–2.3 s, confirming that inference dominates total response time. Infrastructure costs are approximately \$504/month, while API usage contributes an additional ∼\$558/month with GPT-4o under a balanced workload, yielding a total of about \$1,062/month.

Although experiments used a single EC2 instance with Docker Compose for reproducibility, the architecture can be migrated to cloud-native services. PostgreSQL can run on Amazon RDS, Qdrant on ECS/EKS or memory-optimized nodes, Nginx on Fargate or Elastic Beanstalk, and NATS on Amazon MQ, enabling elasticity, fault tolerance, and simplified management in production.

DATA AND CODE AVAILABILITY

The complete implementation of the proposed system and all datasets used for the trainings are available at https://github.com/NahumFGz/tesismaestriauni-launcher. The repository is structured as a main launcher project that integrates several Git submodules, each corresponding to a key component of the architecture (e.g., web interface, microservices, domain-specific MCP agents, data processing pipelines, and evaluation scripts). To ensure full replicability, a root-level *docker-compose.yml* file is included, allowing the entire project to be deployed and executed in a single step. This guarantees that all services are reproduced consistently across environments, thereby facilitating verification and extension of the results. Table 20 summarizes the main modules and their functions.

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